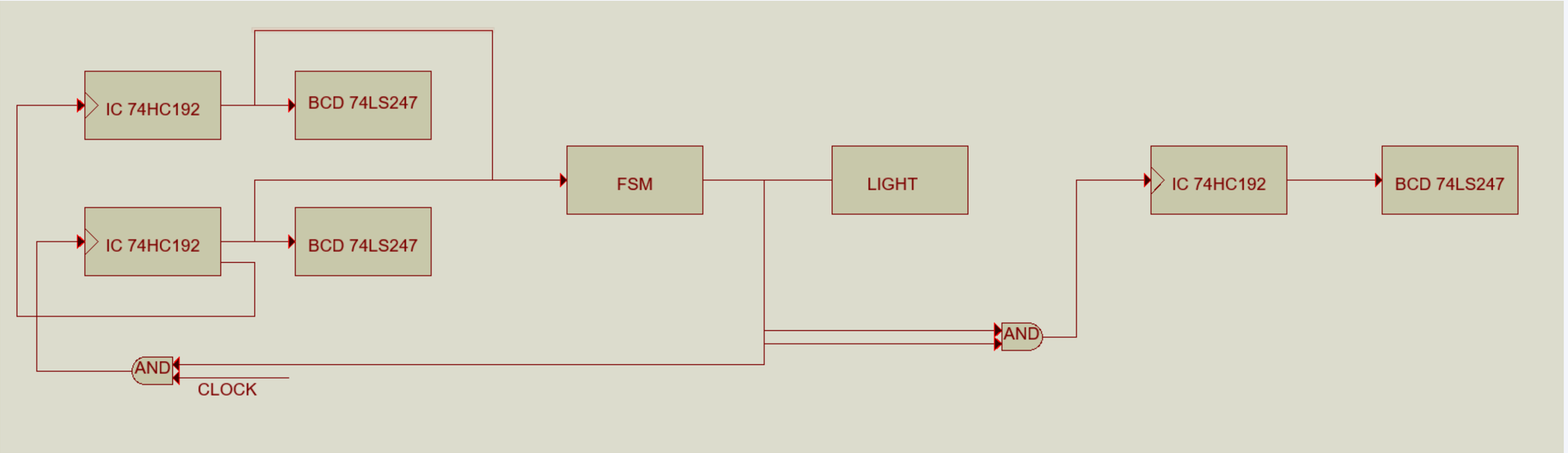
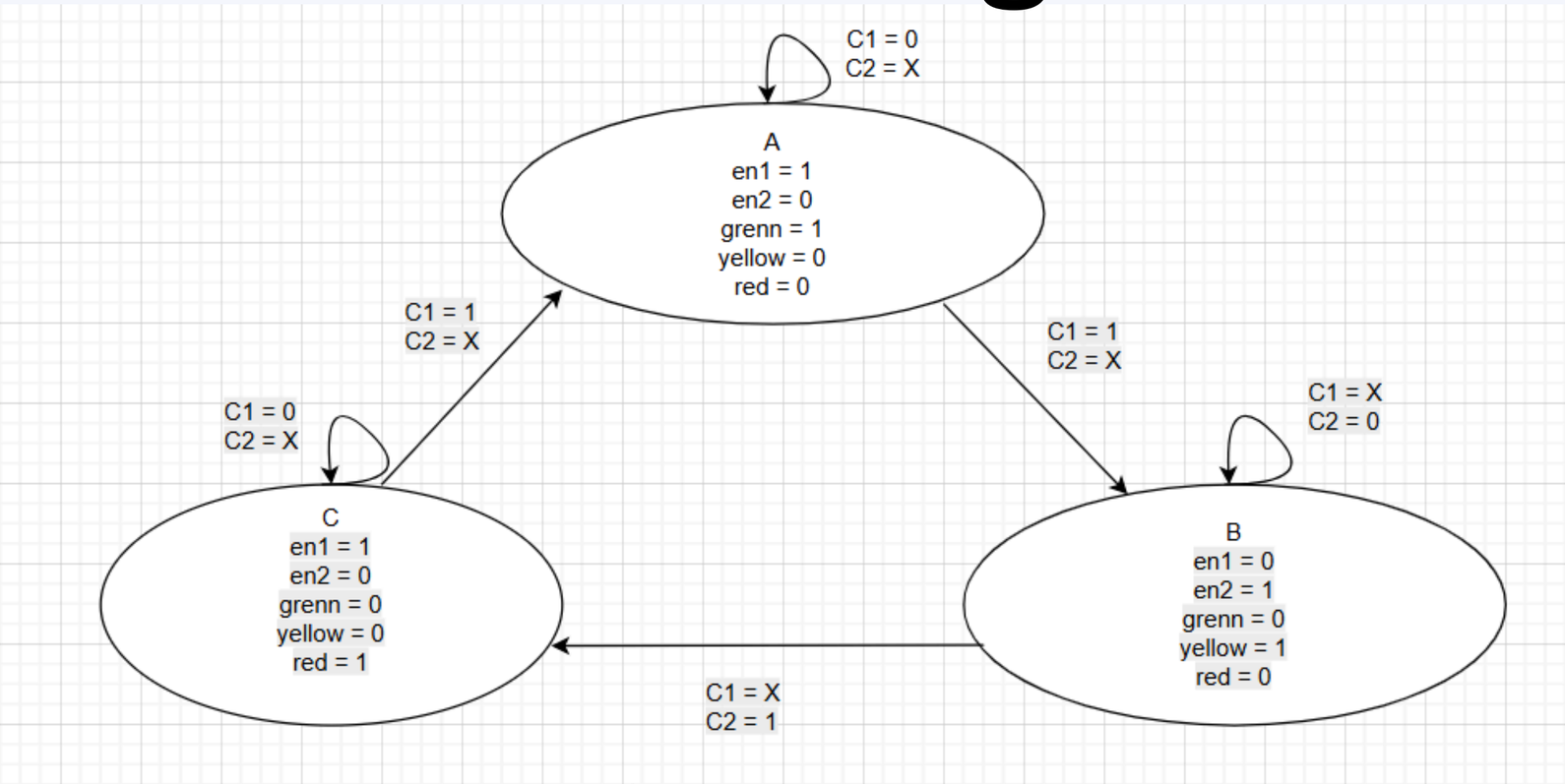
BLOCK DIAGRAM



Sates diagram



Note: C1 = $(Q_3 + Q_2 + Q_1 + Q_0)'$ (the counter 1 for Red and Green light)

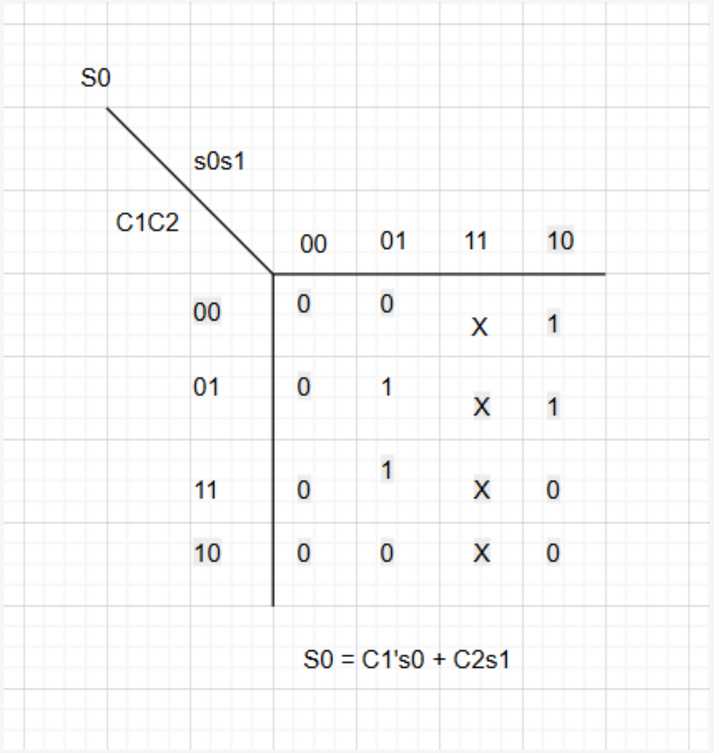
C2 = $(Q_3 + Q_2 + Q_1 + Q_0)'$ (the counter 2 for Yellow light)

State table

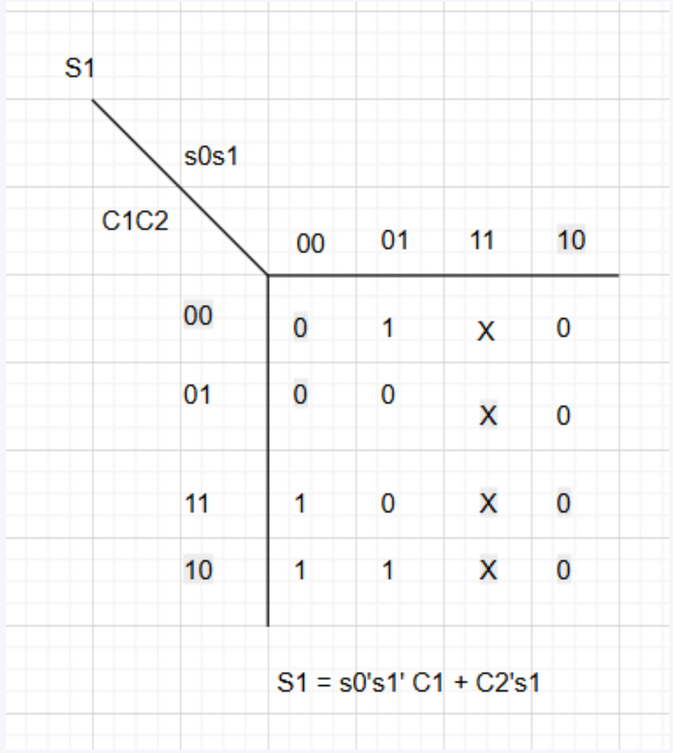
Present stage	New stage (S_0S_1)					Output				
s_0s_1	C_1C_2	00	01	10	11	en_1	en_2	Green	Yellow	Red
A		A	A	B	B	1	0	1	0	0
B		B	C	B	C	0	1	0	1	0
C		C	C	A	A	1	0	0	0	1

State-assigned table

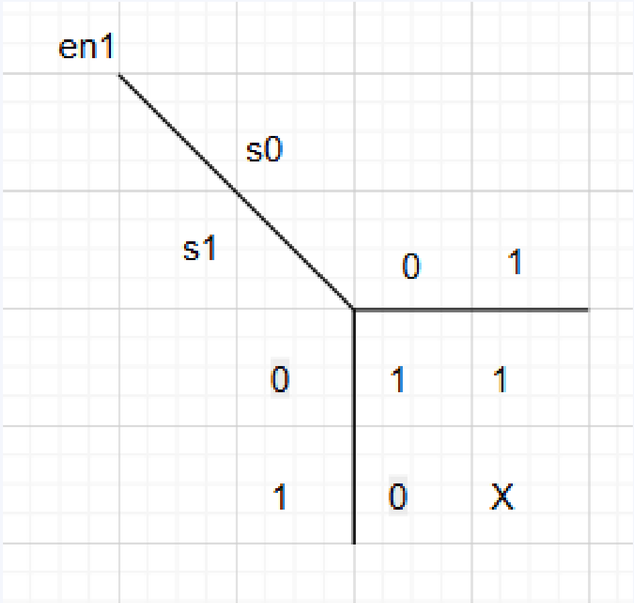
Present stage	New stage (S_0S_1)					Output				
s_0s_1	C_1C_2	00	01	10	11	en_1	en_2	Green	Yellow	Red
00		00	00	01	01	1	0	1	0	0
01		01	10	01	10	0	1	0	1	0
10		10	10	00	00	1	0	0	0	1



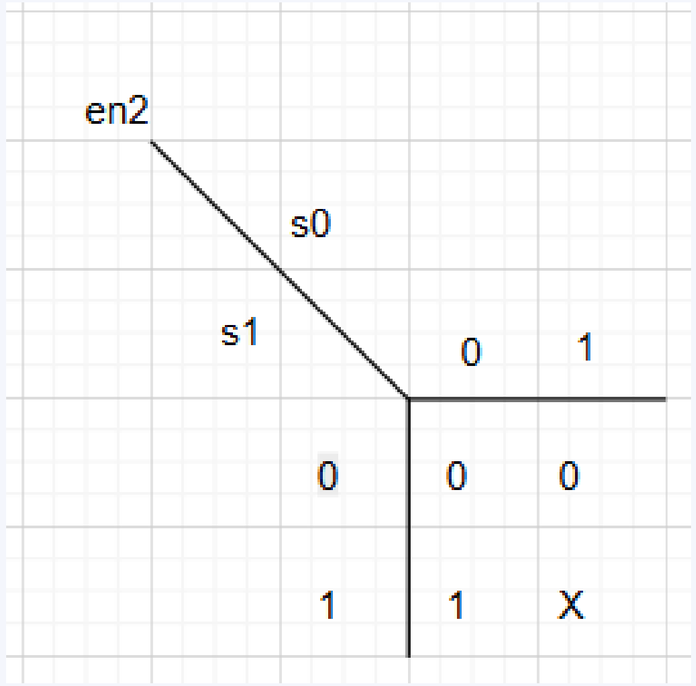
S0 = C1's0 + C2s1



S1 = s0's1' C1 + C2's1



en1 = s1'

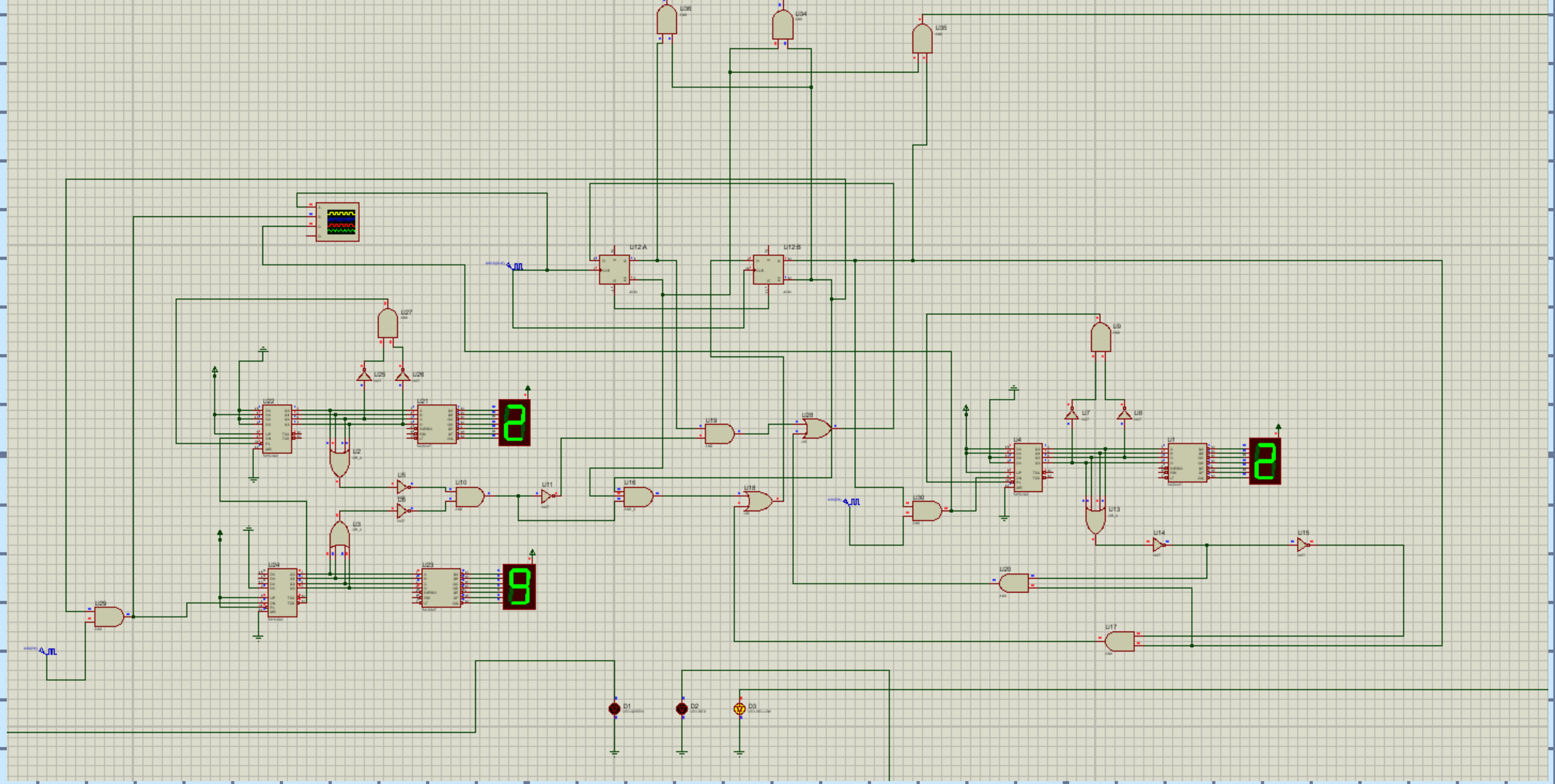


en2 = s1

Red = s0s1'

Green = s0's1'

Yellow = s0's1



Circuit and how it works